

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

2. Supervisor: Dr. Name Surname
Name of Research Centre or Department
University *of* Limerick
Ireland

1. Reviewer: Dr. Name Surname
Name of Research Centre or Department
University of
Country

2. Reviewer: Dr. Name Surname
Name of Research Centre or Department
University of
Country

1. Chair: Dr. Name Surname
Name of Research Centre or Department
University *of* Limerick
Ireland

Day of the defense: day month year

Signature from head of PhD committee:

Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well-organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

Contents

List of Tables	v
List of Figures	vii
List of Abbreviations	x
1 Introduction	1
1.1 Background and Context	1
1.2 Research Problem and Motivation	3
1.2.1 Limited Integration of Multidimensional Data	3
1.2.2 Underuse of Longitudinal Dynamics	3
1.2.3 Lack of Explainability	3
1.3 Research Aims and Objectives	3
1.4 Contributions of the Research	4
1.5 Thesis Outline	4
2 Literature Review	7
2.1 Introduction	7
2.2 Chronic Disease and Ageing	7
2.2.1 Frailty and Physical Function	7
2.2.2 Mental Health in Older Adults	8
2.2.3 Nutrition and Biological Ageing	9
2.3 Machine Learning in Ageing and Chronic Disease	10
2.3.1 Feature Sets in ML Studies	10
2.3.2 ML Models Used	10
2.3.3 Performance and Validation	11

CONTENTS

2.3.4	Explainability in ML for Ageing	11
2.4	Gaps in the Literature	12
2.5	Chapter Summary	12
3	Data and Preprocessing	13
3.1	Insert a Figure	13
3.2	Creating a list	14
3.2.1	Creating a Table	14
3.3	Quote	15
4	Methodology	17
4.1	Math	17
5	Results	19
5.1	References	19
6	Conclusions and Future Directions	21
6.1	Summary	21
6.2	Conclusions	21
6.3	Contributions	21
6.4	Future Work	21
Appendix A: Insert figure in Appendix		23
Appendix B: Insert Text in Appendix		25
Appendix C: Insert C++/Java et al. Code in Appendix		29
References		31

List of Tables

3.1	Table Title	14
-----	-----------------------	----

LIST OF TABLES

List of Figures

3.1	Title of Figure	13
-----	---------------------------	----

LIST OF FIGURES

List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
CES-D	Center for Epidemiologic Studies Depression Scale
DNN	Deep Neural Network
ELSA	The English Longitudinal Study of Aging
EU	European Union
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HRS	The Health and Retirement Study, (a longitudinal study of ageing in the United States)
IDS	The Intellectual Disability Supplement
ISSDA	The Irish Social Science Data Archive
LDA	Linear Discriminant Analysis
ML	Machine Learning
RF	Random Forest
SHAP	Shapley Additive Explanations
SHARE	The Survey of Health, Aging and Retirement in Europe
TILDA	The Irish Longitudinal Study on Ageing

List of Abbreviations

WHO World Health Organization

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions—including cardiovascular disease, diabetes, and frailty—are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non-communicable diseases, contributing billions in healthcare costs annually [6]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [7]. Recognising this, international and national policies—including the World Health Organization (WHO) Global Action Plan on Noncommunicable Diseases and Ireland’s *Healthy Ireland* framework—highlights nutrition and lifestyle interventions as central to healthcare strategy.

In parallel, (Artificial Intelligence (AI)) and (Machine Learning (ML)) are increasingly applied in healthcare to support risk prediction and personalised intervention. However, European regulations such as the General Data Protection Regulation (GDPR) and the forthcoming European Union (EU) AI Act impose strict requirements for transparency, explainability, and governance. This regulatory context underscores the importance of interpretable and ethically robust ML approaches.

1. INTRODUCTION

Traditional risk prediction models have largely relied on clinical biomarkers such as blood pressure and lipid profiles. Yet, growing evidence shows that broader predictors—including nutrition [8], physical activity [9], mental health [10], and social determinants [11]—enhance predictive performance.

(The Irish Longitudinal Study on Ageing (TILDA))¹ provides a nationally representative cohort of adults aged 50+ in Ireland, surveyed across multiple domains in repeated waves (waves 1–5). The raw wave datasets capture the full richness of TILDA-specific measures relevant to Irish health, policy, and society. In parallel, the Harmonized TILDA dataset restructures selected variables to align with the RAND (The Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework, enabling cross-national comparison with equivalent studies such as HRS (US), The English Longitudinal Study of Aging (ELSA) (UK), The Survey of Health, Aging and Retirement in Europe (SHARE) (continental Europe). This harmonisation enhances international comparability, though at the cost of excluding some study-specific variables. An additional dataset, the The Intellectual Disability Supplement (IDS), extends coverage to individuals with disabilities, improving inclusivity and reducing sampling bias. Together, these resources offer both detailed national data and a platform for internationally comparable, generalisable analyses.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over-reliance on clinical predictors, limited incorporation of lifestyle and social variables, predominantly cross-sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by integrating TILDA’s longitudinal, multidomain data with interpretable ML methods, demonstrating how nutrition, physical activity, and mental health can be combined with traditional clinical markers to improve prediction of ageing-related outcomes.

¹TILDA data are pseudonymised and accessed under licence via The Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data governance requirements.

1.2 Research Problem and Motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain constrained in several ways.

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables while underutilising modifiable lifestyle and nutrition factors [8, 12].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [13].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [14].

Addressing these challenges is essential to improve predictive accuracy, support evidence-based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.3 Research Aims and Objectives

This thesis develops and validates ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes in a representative cohort of older adults.

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Preprocess and harmonise TILDA waves 1-5, and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).

1. INTRODUCTION

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how nutrition, lifestyle, and mental health features enhance prediction when combined with established clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML workflows.
- **Clinical:** Identifies modifiable predictors with relevance for targeted interventions and policy to promote healthy ageing.

1.5 Thesis Outline

- *Chapter Two:* Literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations.

- *Chapter Three:* TILDA study design, data preprocessing, harmonisation, and feature engineering.(Flow Charts..)
- *Chapter Four:* Modelling framework, including baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics.(Code snippets..)
- *Chapter Five:* Results, model comparison, feature importance, and ablation analyses.
- *Chapter Six:* Discussion of implications, limitations, ethical considerations, and future directions.

Supporting documents are included in the appendix:

- *Appendix A:* Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B:* Ethics.
- *Appendix C:* Modelling details, supplementary tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive accompanies this dissertation (access request required):

- *Folder 1:* Preprocessing and feature-engineering scripts (documentation only, no raw TILDA data).
- *Folder 2:* Model training/evaluation code and configuration files to reproduce reported results.

1. INTRODUCTION

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA. The focus is on three domains central to later-life health: frailty and physical function, mental health, and nutrition and biological ageing. The review also considers how machine learning has been applied to these areas and identifies gaps that motivate the present study.

2.2 Chronic Disease and Ageing

Chronic conditions are leading causes of morbidity and reduced quality of life in older adults. Multimorbidity, frailty, and mental health difficulties frequently overlap, accelerating functional decline. In line with the WHO healthy ageing framework, research using TILDA highlights frailty, mental health, and nutrition as interconnected determinants of later-life outcomes. The following subsections examine each domain in turn.

2.2.1 Frailty and Physical Function

Frailty reflects declining physiological reserve and heightened vulnerability. In TILDA, it is captured through both the frailty phenotype and the frailty index,

2. LITERATURE REVIEW

with gait speed, grip strength, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

Evidence suggests frailty is dynamic, not irreversible. For example, multi-state modelling showed that many prefrail adults transition back to non-frail, though established frailty strongly predicts mortality [15]. By contrast, metabolic syndrome accelerates frailty onset, pointing to a metabolic pathway of vulnerability [16]. A further nuance is age-dependence: obesity predicts mobility decline earlier in life, while later in life protective factors like grip strength become more important [17]. These findings partly conflict: frailty appears both reversible and progressive, depending on risk context.

Limitations remain. Observational designs constrain causal inference, and predictors are often studied in isolation, ignoring interactions. Few models have been validated beyond the Irish context, raising questions of generalisability. These gaps point to the need for multidimensional approaches. Traditional regression models struggle with reversibility and age-dependent interactions, suggesting a role for machine learning (see Chapter 4).

2.2.2 Mental Health in Older Adults

Mental health, including depression, anxiety, loneliness, and social isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and social factors. [18] found that participation in physical, social, and cognitive activities—summarised by the Act-Belong-Commit (ABC) indicator—was associated with lower depression and anxiety. This suggests social integration and lifestyle protect mental wellbeing. However, citejuolaO117NursingStaff2014 reported that self-reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of social engagement is strong when self-reported, weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer-based measures show a dose-response link between moderate-to-vigorous activity and lower depressive symptoms [9]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, mediated by fear of falling [19].

Together, these results reveal a bidirectional relationship—mental health both influences and is influenced by physical decline.

Most studies rely on broad self-report scales such as Center for Epidemiologic Studies Depression Scale (CES-D) or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation. Samples remain heavily Irish and 50+ in age relevance, limiting wider relevance. This literature suggests mental health in ageing cannot be modelled as a one-way effect. It requires longitudinal, multidimensional approaches that can capture reciprocal pathways—an area where machine learning is well suited.

2.2.3 Nutrition and Biological Ageing

Nutrition is increasingly recognised as a modifiable factor in ageing and frailty. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [8] showed that higher plasma lutein and zeaxanthin—dietary carotenoids found in leafy vegetables—were linked to slower frailty progression. By contrast, [20] found vitamin D deficiency predicted diabetes and musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

Malnutrition also reflects social and functional risks. [2] found that hospitalisation, functional loss, and poor social support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating social context.

A central debate is the distinction between chronological and biological age. Biomarker-based measures such as epigenetic clocks or frailty indices often diverge from chronological age, better capturing this cumulative exposures. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [21, 22]. These measures provide finer-grained prediction of vulnerability, but most evidence is cross-sectional, with limited power to capture changes over time. Small sample sizes for biomarker studies also reduce reliability.

Taken together, nutrition interacts with biological, metabolic, and social systems. Isolating single nutrients risks oversimplification. Integrative modelling

2. LITERATURE REVIEW

that combines biomarkers, clinical factors, and social determinants—supported by machine learning pipelines—offers a more promising route.

2.3 Machine Learning in Ageing and Chronic Disease

ML is increasingly applied in ageing research because it can capture non-linear and multidimensional relationships that traditional regression models—such as logistic regression—often miss [23, 24].

2.3.1 Feature Sets in ML Studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often emphasize easily measured physical health data—such as comorbidity counts, mobility scores, or vital signs—while more complex but important predictors, including nutrition and social factors, are underrepresented [25]. This imbalance may limit the ability of models to capture the full complexity of ageing, especially considering that mental health and nutrition exert independent and interacting effects on later-life outcomes (see Section 2.2). Models that focus only on readily available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.3.2 ML Models Used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non-linearities or higher-order interactions. More recent work has shifted towards ensemble methods such as random forests (RF) and gradient boosting machines (GBM), as well as deep neural networks (Deep Neural Network (DNN)), which generally achieve higher predictive accuracy.

Similarly, [26] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an AUC of 0.78. By contrast, ensemble methods in [23] reached 0.854 for diabetes prediction, illustrating how flexible, non-linear models can better capture complex interactions among ageing-related predictors.

2.3.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like diabetes, gait speed, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. Additionally, most research is cross-sectional, capturing associations at a single point in time rather than true longitudinal trends, so causal and temporal relationships remain unclear. When combined with the underuse of nutrition and social predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context-specific results rather than generalisable predictions [25].

2.3.4 Explainability in ML for Ageing

As ML models become more complex, interpretability emerges as a critical challenge. Tools such as Shapley Additive Explanations (SHAP) values and TreeExplainer have been used to rank feature importance and provide local explanations [3, 23, 27]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones. Nonetheless, black-box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented,

2. LITERATURE REVIEW

with no consensus on best practice. This gap links directly to the broader methodological limitations discussed in the next section on gaps in the literature.

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological
- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
 - integrating nutrition + lifestyle + clinical predictors
 - ML + explainability
 - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).



Figure 3.1: Title of Figure - Description. [Source: (?)]

3. DATA AND PREPROCESSING

3.2 Creating a list

- (2000) item 1.
- (2004) item 2.
- (2010) item 3.
- (2013) item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

*There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.*

(? , verses 208-218)

3. DATA AND PREPROCESSING

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol T denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.5}$$

4. METHODOLOGY

$$\begin{aligned}
&= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{|A_{ba}| |M_{bm}| \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}} \\
&= \frac{-0.2338, -0.0931, -0.0651}{0.2601} \\
&= (-0.8989, -0.3579, -0.2503)
\end{aligned}$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file ‘references_myphd’ with BibTex. The file is located in ‘/7_references/’

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book:** (? , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter:** (?)
- **Book Contribution:** (?)
- **MUSIC CD/DVD:** (?)
- **Journal Article:** (?)
- **Patent:** (?)
- **PhD/Master Dissertation:** (?)
- **Conference Paper:** (?)
- **Technical Report:** (?)

5. RESULTS

- **Unpublished Report:** (?)
- **Web Videos**(?) (. . . not the name of the person who uploaded the video though)
- **Web Paper:** (?)
- **Web Journal Article:** (?)
- **Webpage:** (?)
- **Downloaded Images or Sounds:** (?)

You can also use this kind of referencing if needed in the text:
Surname [?] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

6. CONCLUSIONS AND FUTURE DIRECTIONS

Appendix A

Insert a figure

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 * Copyright (c) 2013, Giuseppe Torre
 * All rights reserved.
 *
 * Redistribution and use in source and binary forms, with or without
 * modification, are permitted provided that the following conditions are met:
 *     * Redistributions of source code must retain the above copyright
 *       notice, this list of conditions and the following disclaimer.
 *     * Redistributions in binary form must reproduce the above copyright
 *       notice, this list of conditions and the following disclaimer in the
 *       documentation and/or other materials provided with the distribution.
 *     * Neither the name of the BREAKFAST LLC nor the
 *       names of its contributors may be used to endorse or promote products
 *       derived from this software without specific prior written permission.
 *
 * THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND
 * ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED
 * WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE
 * DISCLAIMED. IN NO EVENT SHALL BREAKFAST LLC BE LIABLE FOR ANY
 * DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES
 * (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 * LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 * ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 * (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 * SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 */

#include "ext.h"
#include "ext.mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//*****CALIBRATION STUFF*****//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX.OFFSET 2043 //initial offset in accelerometer X
#define ACCELY.OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ.OFFSET 2264//initial offset in accelerometer Z

#define MagX.OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667 // initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
// ***** END CALIBRATION STUFF *****//

void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
    t_object m_ob; // Must always be the first field; used by Max

    Atom m_args[9]; // we want our inlet to be receiving a list of 10 elements

    long m_value; //inlet

    void *m_R1; //these are all the outlets for the 3 X 3 Matrix
    void *m_R2; // R1 -> firts top left cell ..R2 middle cell of the first
    row ...and so on
    void *m_R3; //
    void *m_R4; // End Outlets
} t_triad;

void *triad_new(long value);

void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
s);
void triad_free(t_triad *triad);

void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);

void MatrixByMatrix(double *Result, double *MatrixLeft, double *
MatrixRight);

void Matrix2Quat(double *Quat, double *Matrix);
void Quat2Matrix(double *Matrix, double *Quat);
void inverseQuat(double *InvQuat, double *RegQuat);
void NormQuat(double *YesQuat, double *NotQuat);
void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
void NormVect(double *YesVect, double *NotVect);
double orientationMatrix[9];
double Result[9];
int i;
double InvorientationMatrix[9];

double temp[6];
double ref[6];
double vectAx[3];
double vectAy[3];
double vectAz[3];
double vectBx[3];
double vectBy[3];
double vectBz[3];
double MagnCrosProd_A;
double MagnCrosProd_B;
double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
VectBynorm, VectBznorm;
```

```

        double m[9], n[9];
        double quat_e[4];
        double invquat_e[4];
        double mult;
        double quat_new[4];
        double quat_old[4];
        double ecs, accex_ADCnumber, y, accey_ADCnumber, z, accez_ADCnumber, mx,
            magnx_ADCnumber, my, magny_ADCnumber, mz, magnz_ADCnumber;

// SLERP Variables

        double trace, Suca;
        double tol[4], omega, sinom, cosom, scale0, scale1, tez,
            orientationMatrixA[9];

int main(void)
{
    // set up our class: create a class definition
    setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
        sizeof(t_triad), 0L, A_GIMME, 0);
    address((method)triad_list, "list", A_GIMME, 0);
    address((method)triad_assist, "assist", A_CANT, 0);
    finder_addclass("Maths", "triad");
    post(".... I 'm-Triad-Object !.... from -AHRS- Library ...", 0);
    return 0;
}

/* ----- triad_new ----- */

void *triad_new(long value)
{
    t_triad *triad;
    triad = (t_triad *)newobject(this_class);           // create the new instance and return
        a pointer to it
    triad->m_R4 = floatout(triad);
    triad->m_R3 = floatout(triad);
    triad->m_R2 = floatout(triad);
    triad->m_R1 = floatout(triad);

    return(triad);
}

etc.

etc.

```

Appendix

References

- [1] Leonardo Bencivenga, Klara Komici, Pierangela Nocella, Fabrizio Vincenzo Grieco, Angela Spezzano, Brunella Puzone, Alessandro Cannavo, Antonio Cittadini, Graziamaria Corbi, Nicola Ferrara, and Giuseppe Rengo. Atrial fibrillation in the elderly: A risk factor beyond stroke. *Ageing Research Reviews*, 61:101092, 2020. ISSN 1568-1637. doi: 10.1016/j.arr.2020.101092. 1
- [2] Laura A Bardon, Melanie Streicher, Clare A Corish, Michelle Clarke, Lauren C Power, Rose Anne Kenny, Deirdre M O'Connor, Eamon Laird, Eibhlis M O'Connor, Marjolein Visser, Dorothee Volkert, Eileen R Gibney, and MaNuEL Consortium. Predictors of Incident Malnutrition in Older Irish Adults from the Irish Longitudinal Study on Ageing Cohort—A MaNuEL study. *J Gerontol A Biol Sci Med Sci*, 75(2):249–256, January 2020. ISSN 1079-5006. doi: 10.1093/gerona/gly225. 1, 9
- [3] James R. C. Davis, Silvin P. Knight, Orna A. Donoghue, Belinda Hernández, Rossella Rizzo, Rose Anne Kenny, and Roman Romero-Ortuno. Comparison of Gait Speed Reserve, Usual Gait Speed, and Maximum Gait Speed of Adults Aged 50+ in Ireland Using Explainable Machine Learning. *Front Netw Physiol*, 1:754477, 2021. ISSN 2674-0109. doi: 10.3389/fnetp.2021.754477. 1, 11
- [4] Patrick V. Moore, Kathleen Bennett, and Charles Normand. Counting the time lived, the time left or illness? Age, proximity to death, morbidity and prescribing expenditures. *Social Science & Medicine*, 184:1–14, 2017. ISSN 0277-9536. doi: 10.1016/j.socscimed.2017.04.038. 1
- [5] Ying Huang, Yuhao Su, Ying Jiang, and Meilan Zhu. Sex differences in the associations between blood pressure and anxiety and depression scores in a middle-aged and elderly population: The Irish Longitudinal Study on Ageing (TILDA). *Jour-*

REFERENCES

- nal of Affective Disorders*, 274:118–125, September 2020. ISSN 0165-0327. doi: 10.1016/j.jad.2020.05.133. 1
- [6] World Health Organization Europe. New data: Noncommunicable diseases cause 1.8 million avoidable deaths and cost US\$ 514 billion every year, reveals new WHO/Europe report. <https://www.who.int/europe/news/item/27-06-2025-new-data-noncommunicable-diseases-cause-1-8-million-avoidable-deaths-and-cost-us-514-billion-USD-every-year-reveals-new-who-europe-report>, 2025. 1
- [7] Niamh Rice and Charles Normand. The cost associated with disease-related malnutrition in Ireland. *Public Health Nutr*, 15(10):1966–1972, October 2012. ISSN 1475-2727. doi: 10.1017/S1368980011003624. 1
- [8] Caoileann H. Murphy, Eoin Duggan, James Davis, Aisling M. O’Halloran, Silvin P. Knight, Rose Anne Kenny, Sinead N. McCarthy, and Roman Romero-Ortuno. Plasma lutein and zeaxanthin concentrations associated with musculoskeletal health and incident frailty in The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 171:112013, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112013. 2, 3, 9
- [9] Eamon Laird, Charlotte Lund Rasmussen, Rose Anne Kenny, and Matthew P. Herring. Physical Activity Dose and Depression in a Cohort of Older Adults in The Irish Longitudinal Study on Ageing. *JAMA Netw Open*, 6(7):e2322489, July 2023. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2023.22489. 2, 8
- [10] C. P. McDowell, R. K. Dishman, M. Hallgren, C. MacDonncha, and M. P. Herring. Associations of physical activity and depression: Results from the Irish Longitudinal Study on Ageing. *Experimental Gerontology*, 112:68–75, 2018. ISSN 0531-5565. doi: 10.1016/j.exger.2018.09.004. 2
- [11] Cathal McCrory, Ciaran Finucane, Celia O’Hare, John Frewen, Hugh Nolan, Richard Layte, Patricia M. Kearney, and Rose Anne Kenny. Social Disadvantage and Social Isolation Are Associated With a Higher Resting Heart Rate: Evidence From The Irish Longitudinal Study on Ageing. *J Gerontol B Psychol Sci Soc Sci*, 71(3):463–473, May 2016. ISSN 1079-5014. doi: 10.1093/geronb/gbu163. 2
- [12] Eamon Laird, Matthew P. Herring, Brian P. Carson, Catherine B. Woods, Cathal Walsh, Rose Anne Kenny, and Charlotte Lund Rasmussen. Physical activity for

REFERENCES

- depression among the chronically ill: Results from older diabetics in the Irish longitudinal study on ageing. *Psychiatry Research*, 326:115274, August 2023. ISSN 0165-1781. doi: 10.1016/j.psychres.2023.115274. 3
- [13] Sebastian Moguilner, Silvin P. Knight, James R. C. Davis, Aisling M. O’Halloran, Rose Anne Kenny, and Roman Romero-Ortuno. The Importance of Age in the Prediction of Mortality by a Frailty Index: A Machine Learning Approach in the Irish Longitudinal Study on Ageing. *Geriatrics (Basel)*, 6(3):84, August 2021. ISSN 2308-3417. doi: 10.3390/geriatrics6030084. 3
- [14] Orna A Donoghue, Belinda Hernandez, Matthew D L O’Connell, and Rose Anne Kenny. Using Conditional Inference Forests to Examine Predictive Ability for Future Falls and Syncope in Older Adults: Results from The Irish Longitudinal Study on Ageing. *J Gerontol A Biol Sci Med Sci*, 78(4):673–682, April 2023. ISSN 1758-535X. doi: 10.1093/gerona/glac156. 3
- [15] Roman Romero-Ortuno, Peter Hartley, James Davis, Silvin P. Knight, Rossella Rizzo, Belinda Hernández, Rose Anne Kenny, and Aisling M. O’Halloran. Transitions in frailty phenotype states and components over 8 years: Evidence from The Irish Longitudinal Study on Ageing. *Archives of Gerontology and Geriatrics*, 95: 104401, 2021. ISSN 0167-4943. doi: 10.1016/j.archger.2021.104401. 8
- [16] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Padraic Fallon, Román Romero Ortuño, and Rose Anne Kenny. Association between metabolic syndrome and risk of both prevalent and incident frailty in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 172:112056, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112056. 8
- [17] Chuanying Huang, Shuqin Sun, Xiaocao Tian, Tong Wang, Tianyu Wang, Haiping Duan, and Yili Wu. Age modify the associations of obesity, physical activity, vision and grip strength with functional mobility in Irish aged 50 and older. *Archives of Gerontology and Geriatrics*, 84:103895, 2019. ISSN 0167-4943. doi: 10.1016/j.archger.2019.05.020. 8
- [18] Ziggi Ivan Santini, Ai Koyanagi, Stefanos Tyrovolas, Josep Maria Haro, Robert J. Donovan, Line Nielsen, and Vibeke Koushede. The protective properties of Act-Belong-Commit indicators against incident depression, anxiety, and cognitive impairment among older Irish adults: Findings from a prospective community-

REFERENCES

- based study. *Experimental Gerontology*, 91:79–87, 2017. ISSN 0531-5565. doi: 10.1016/j.exger.2017.02.074. 8
- [19] Louis Jacob, Karel Kostev, Jae Il Shin, Lee Smith, Hans Oh, Adel S. Abduljabbar, Josep Maria Haro, and Ai Koyanagi. Falls increase the risk for incident anxiety and depressive symptoms among adults aged ≥ 50 years: An analysis of the Irish longitudinal study on ageing. *Archives of Gerontology and Geriatrics*, 114:105098, 2023. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105098. 8
- [20] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Cathal Walsh, Martin Healy, Annette L. Fitzpatrick, James B. Walsh, Belinda Hernández, Padraic Fallon, Anne M. Molloy, and Rose Anne Kenny. Association between vitamin D deficiency and the risk of prevalent type 2 diabetes and incident prediabetes: A prospective cohort study using data from The Irish Longitudinal Study on Ageing (TILDA). *EClinicalMedicine*, 53:101654, November 2022. ISSN 2589-5370. doi: 10.1016/j.eclinm.2022.101654. 9
- [21] Kevin McCarthy, Aisling M. O’Halloran, Padraic Fallon, Rose Anne Kenny, and Cathal McCrory. Metabolic syndrome accelerates epigenetic ageing in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 183:112314, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2023.112314. 9
- [22] Zahra Azizi, Rebecca J. Hirst, Alan O’ Dowd, Cathal McCrory, Rose Anne Kenny, Fiona N. Newell, and Annalisa Setti. Evidence for an association between allostatic load and multisensory integration in middle-aged and older adults. *Archives of Gerontology and Geriatrics*, 116:105155, 2024. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105155. 9
- [23] Xuezhong Xu, Xue Mingyang, Jie Yang, Hailong Zheng, and Zhifei Che. Tailored machine learning for evaluating the long-term diabetes risk in older individuals: Findings from the Irish Longitudinal Study on Ageing (TILDA). *BMJ Open*, 13(5):e072991, May 2023. ISSN 2044-6055. doi: 10.1136/bmjopen-2023-072991. 10, 11
- [24] Rory Boyle, Lee Jollans, Laura M. Rueda-Delgado, Rossella Rizzo, Görsev G. Yener, Jason P. McMorrow, Silvin P. Knight, Daniel Carey, Ian H. Robertson, Derya D. Emek-Savaş, Yaakov Stern, Rose Anne Kenny, and Robert Whelan. Brain-predicted age difference score is related to specific cognitive functions: A

REFERENCES

- multi-site replication analysis. *Brain Imaging Behav*, 15(1):327–345, February 2021. ISSN 1931-7565. doi: 10.1007/s11682-020-00260-3. 10
- [25] A. A. Zarudsky and K. I. Prashchayeu. P277: One-legged balance test and falls in old patients with cardiovascular diseases. *European Geriatric Medicine*, 5:S171–S172, 2014. ISSN 1878-7649. doi: 10.1016/S1878-7649(14)70448-6. 10, 11
- [26] Soraya Matthews, Mark Ward, Anne Nolan, Charles Normand, Rose Anne Kenny, and Peter May. Predicting mortality in The Irish Longitudinal Study on Ageing (TILDA): Development of a four-year index and comparison with international measures. *BMC Geriatr*, 22:510, June 2022. ISSN 1471-2318. doi: 10.1186/s12877-022-03196-z. 11
- [27] James Davis, Silvin P. Knight, Rossella Rizzo, Orna A. Donoghue, Rose Anne Kenny, and Roman Romero-Ortuno. A linear regression-based machine learning pipeline for the discovery of clinically relevant correlates of gait speed reserve from multiple physiological systems. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 1266–1270, August 2021. doi: 10.23919/EUSIPCO54536.2021.9616187. 11